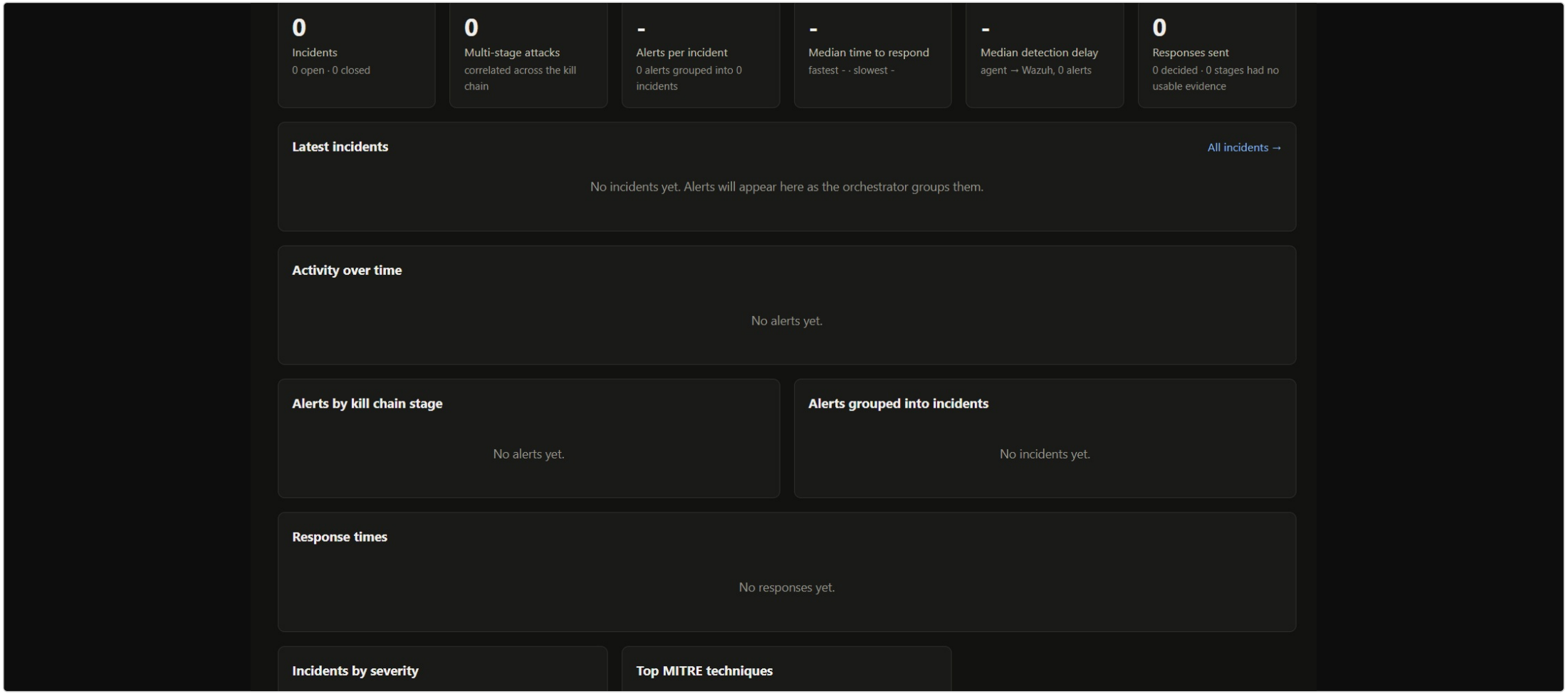
1 OPENING — CLEAN BOARD

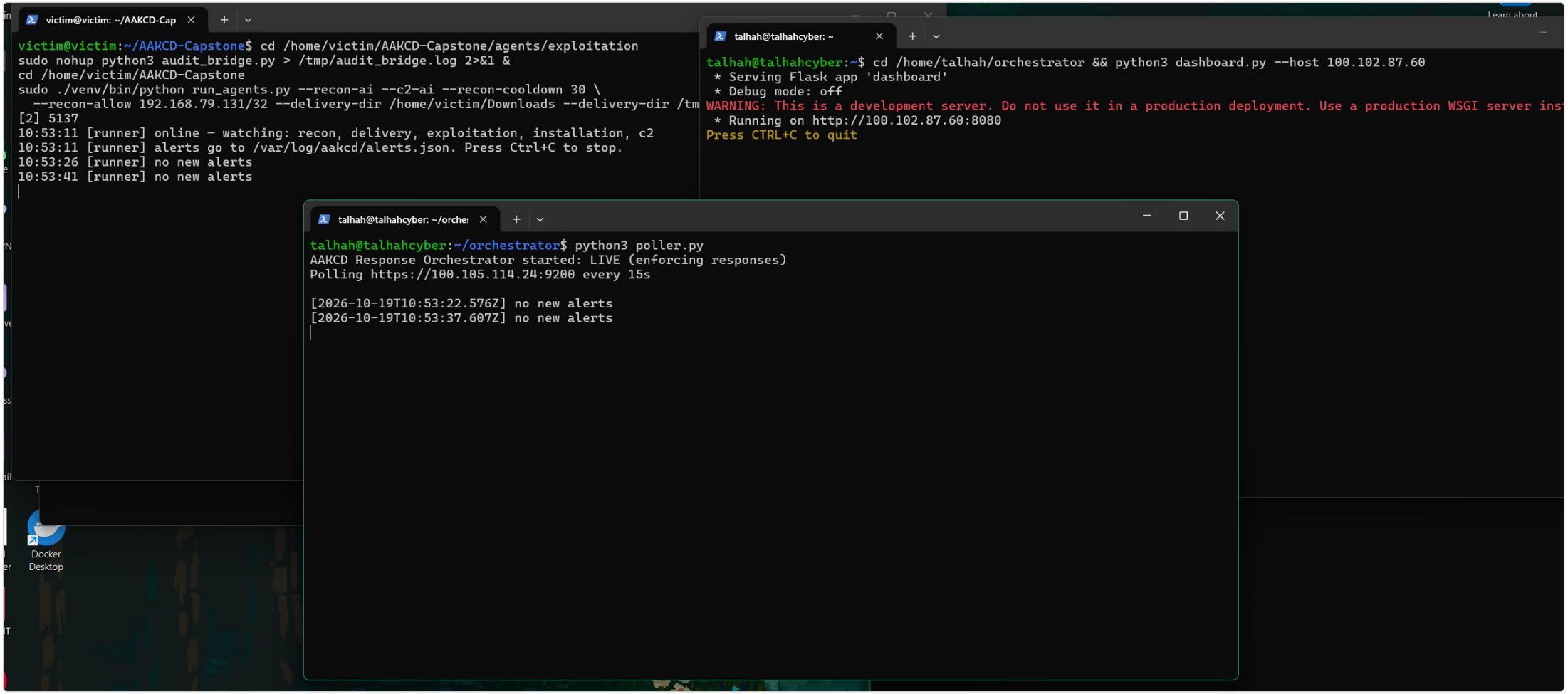
Before the attack. The orchestrator dashboard on a clean slate, every metric zeroed, waiting for the first alert.



c71e3d83-image.png

2OPENING — SYSTEM ONLINE

The defence goes live. Five detection agents watch the victim, the orchestrator polls Wazuh every 15 seconds, the dashboard is up. No attacks yet.



cabb0927-image.png

3 SESSION 1 — RECON — ATTACK

Stage 1: Reconnaissance. From Kali, the attacker SYN-scans 1,000 ports on the victim.

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -p 1-1000 192.168.79.131
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-19 07:14 -0400
Nmap scan report for 192.168.79.131
Host is up (0.0011s latency).
Not shown: 998 closed tcp ports (reset)
PORT    STATE SERVICE
22/tcp  open  ssh
465/tcp open  smtps
MAC Address: 00:0C:29:45:13:C0 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.71 seconds

┌──(kali㉿kali)-[~]
└─$
```

c3941c4a-image.png

4 SESSION 1 — RECON — DETECTED

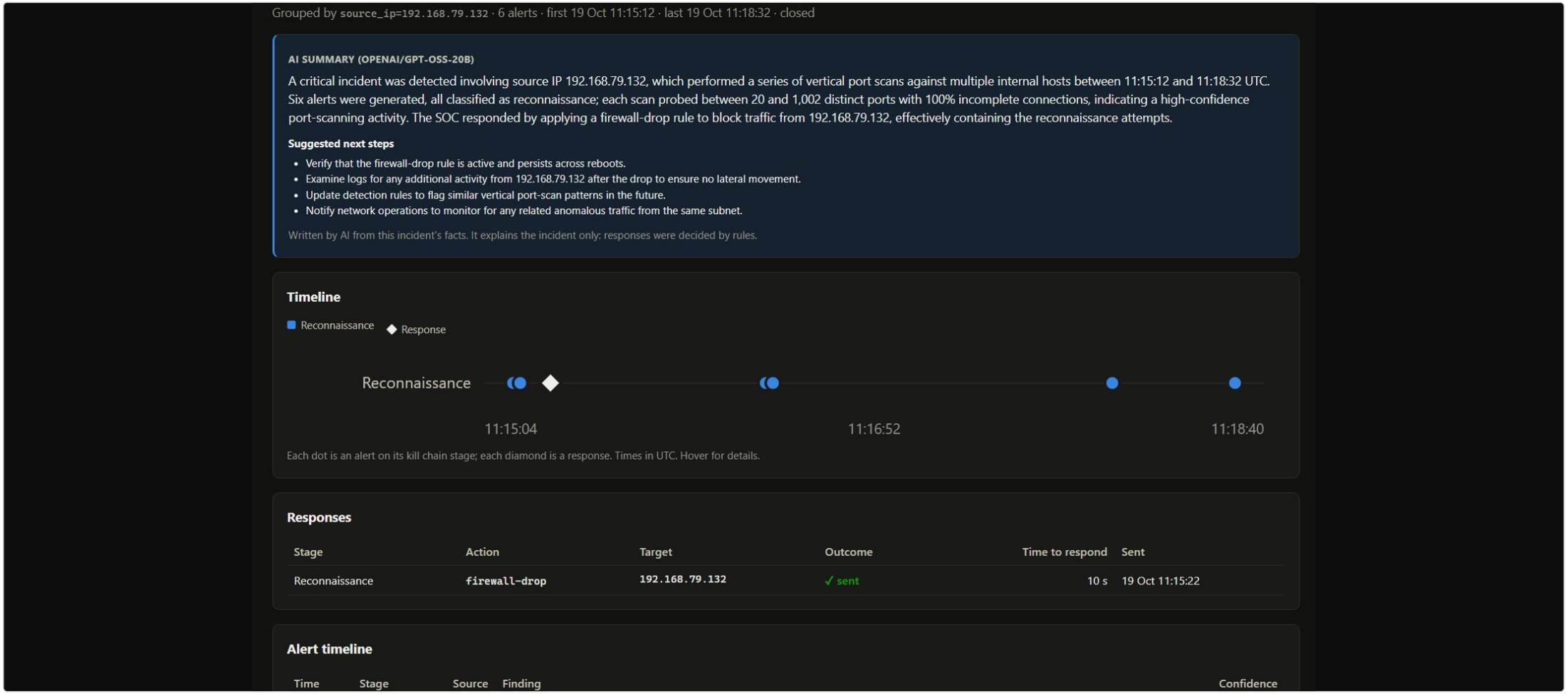
The recon agent catches it live: vertical scans from 192.168.79.132, escalating MEDIUM to HIGH as the scan widens.

```
11:12:23 [runner] no new alerts
11:12:38 [runner] no new alerts
11:12:53 [runner] no new alerts
11:13:08 [runner] no new alerts
11:13:23 [runner] no new alerts
11:13:38 [runner] no new alerts
11:13:53 [runner] no new alerts
11:14:08 [runner] no new alerts
11:14:26 [runner] no new alerts
11:14:39 [runner] no new alerts
11:14:54 [runner] no new alerts
11:15:10 [runner] no new alerts
11:15:11 [recon] MEDIUM vertical scan from 192.168.79.132 (20 targets)
11:15:13 [recon] HIGH vertical scan from 192.168.79.132 (100 targets)
11:15:25 [runner] no new alerts
11:15:41 [runner] no new alerts
11:15:56 [runner] no new alerts
11:16:13 [runner] no new alerts
11:16:21 [recon] MEDIUM vertical scan from 192.168.79.132 (20 targets)
11:16:23 [recon] HIGH vertical scan from 192.168.79.132 (100 targets)
11:16:29 [runner] no new alerts
11:16:46 [runner] no new alerts
11:17:03 [runner] no new alerts
11:17:19 [runner] no new alerts
11:17:36 [delivery] 1 alert(s) for invoice.pdf.exe
11:17:36 [runner] 1 new alert(s) this cycle (see above)
11:17:52 [runner] no new alerts
11:17:58 [recon] HIGH vertical scan from 192.168.79.132 (1002 targets)
```

9b62f991-image.png

5 SESSION 1 — RECON — CONTAINED

The orchestrator correlates the scans into incident #1 and firewall-drops the scanner, 10 seconds after the first alert.



7ba2e8ba-image.png

6 SESSION 1 — RECON — PROOF

Proof it worked: the attacker re-runs the exact scan and every port now comes back filtered. Kali is cut off.

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -p 1-1000 192.168.79.131
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-19 07:14 -0400
Nmap scan report for 192.168.79.131
Host is up (0.0011s latency).
Not shown: 998 closed tcp ports (reset)
PORT    STATE SERVICE
22/tcp  open  ssh
465/tcp open  smtps
MAC Address: 00:0C:29:45:13:C0 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.71 seconds

┌──(kali㉿kali)-[~]
└─$ sudo nmap -sS -p 1-1000 192.168.79.131
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-19 07:16 -0400
Nmap scan report for 192.168.79.131
Host is up (0.00033s latency).
All 1000 scanned ports on 192.168.79.131 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:0C:29:45:13:C0 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 21.80 seconds

┌──(kali㉿kali)-[~]
└─$
```

23790889-image.png

7 SESSION 1 — DELIVERY — ATTACK

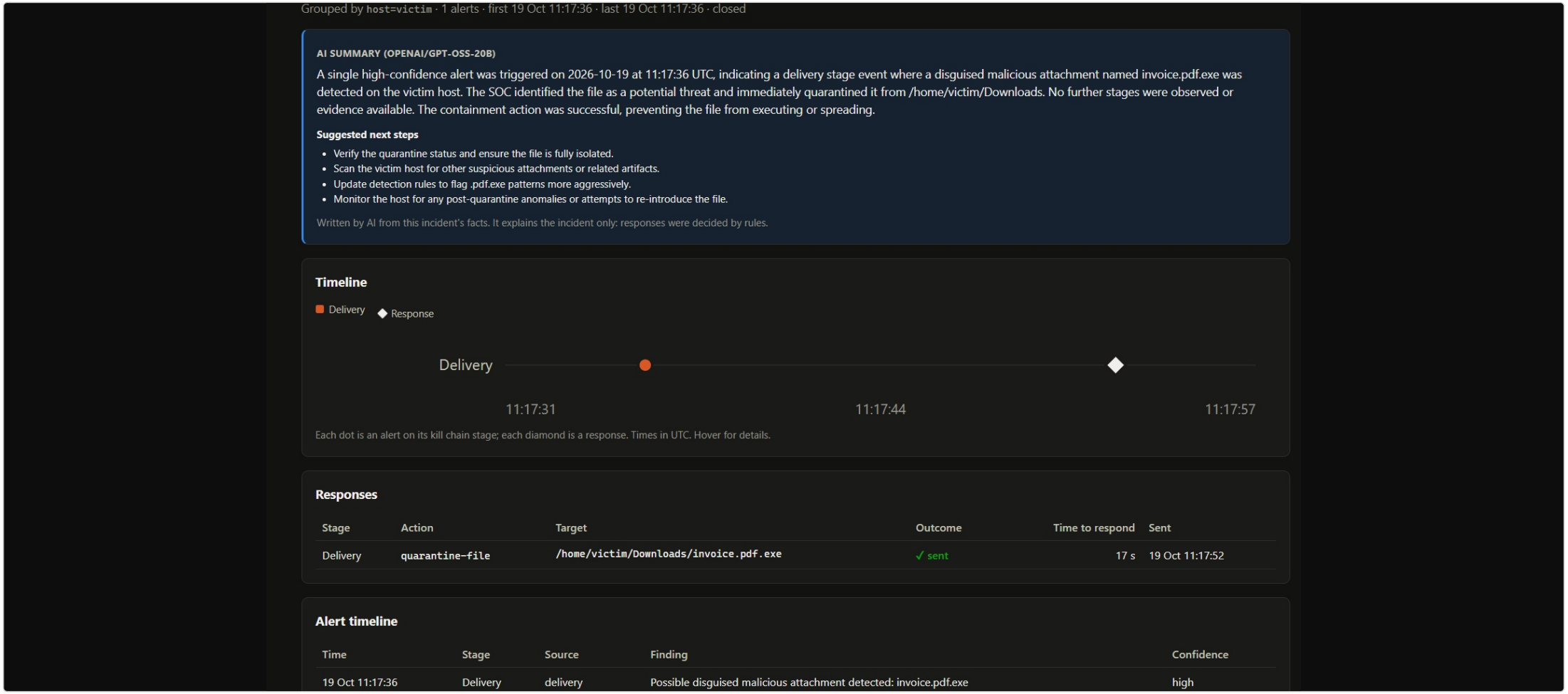
Stage 2: Delivery. A disguised double-extension file (invoice.pdf.exe) drops into the victim's Downloads.

```
victim@victim:~/AAKCD-Capstone$ echo "harmless demo payload for AAKCD delivery stage" > /home/victim/Downloads/invoice.p
df.exe
victim@victim:~/AAKCD-Capstone$
```

9cab3195-image.png

8 SESSION 1 — DELIVERY — CONTAINED

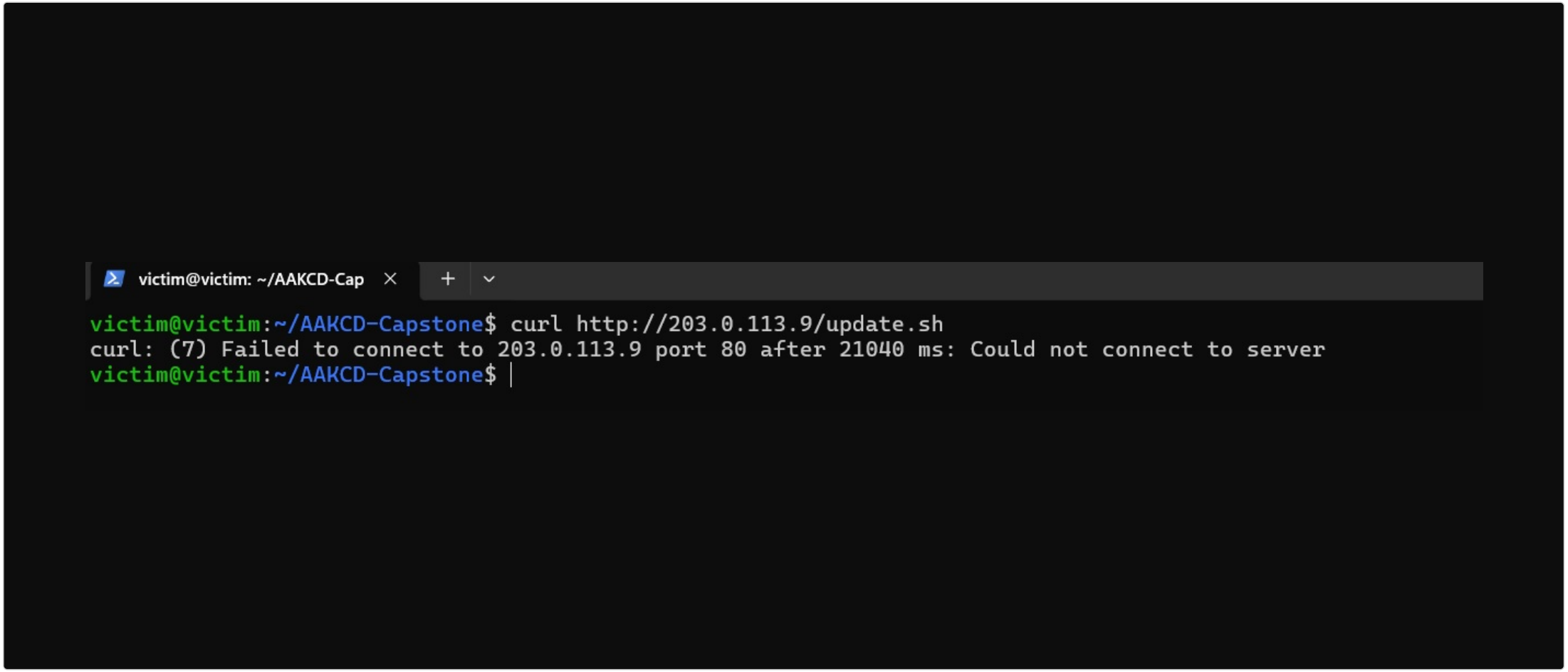
The delivery agent flags it (T1566.001) and the orchestrator quarantines the file from Downloads in 17 seconds.



fa1cdfe7-image.png

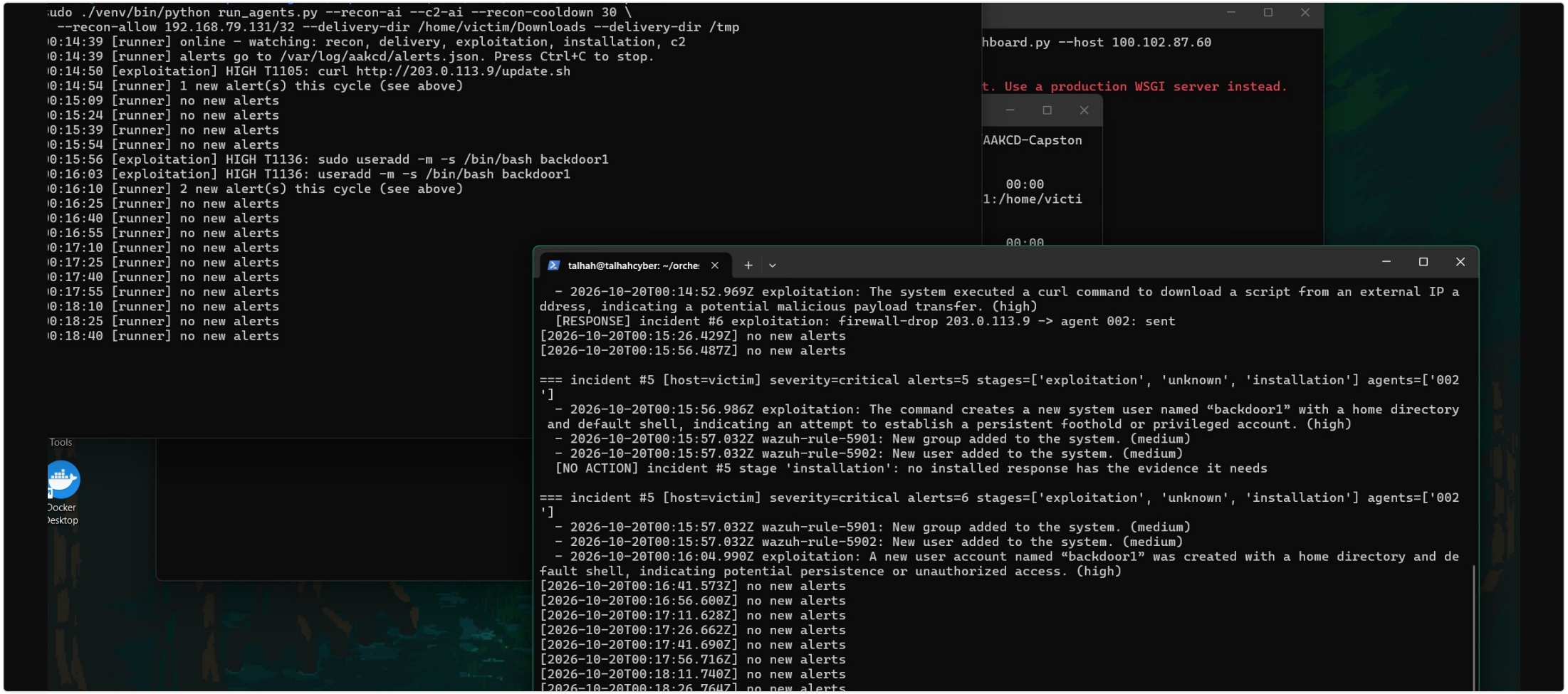
9 SESSION 2 — EXPLOITATION — ATTACK

Stage 3: Exploitation. The attacker runs a download-and-execute command, fetching a script from an external server (T1105). The IP is a safe documentation address; the command itself is what gets caught.



ed1f1f90-image.png

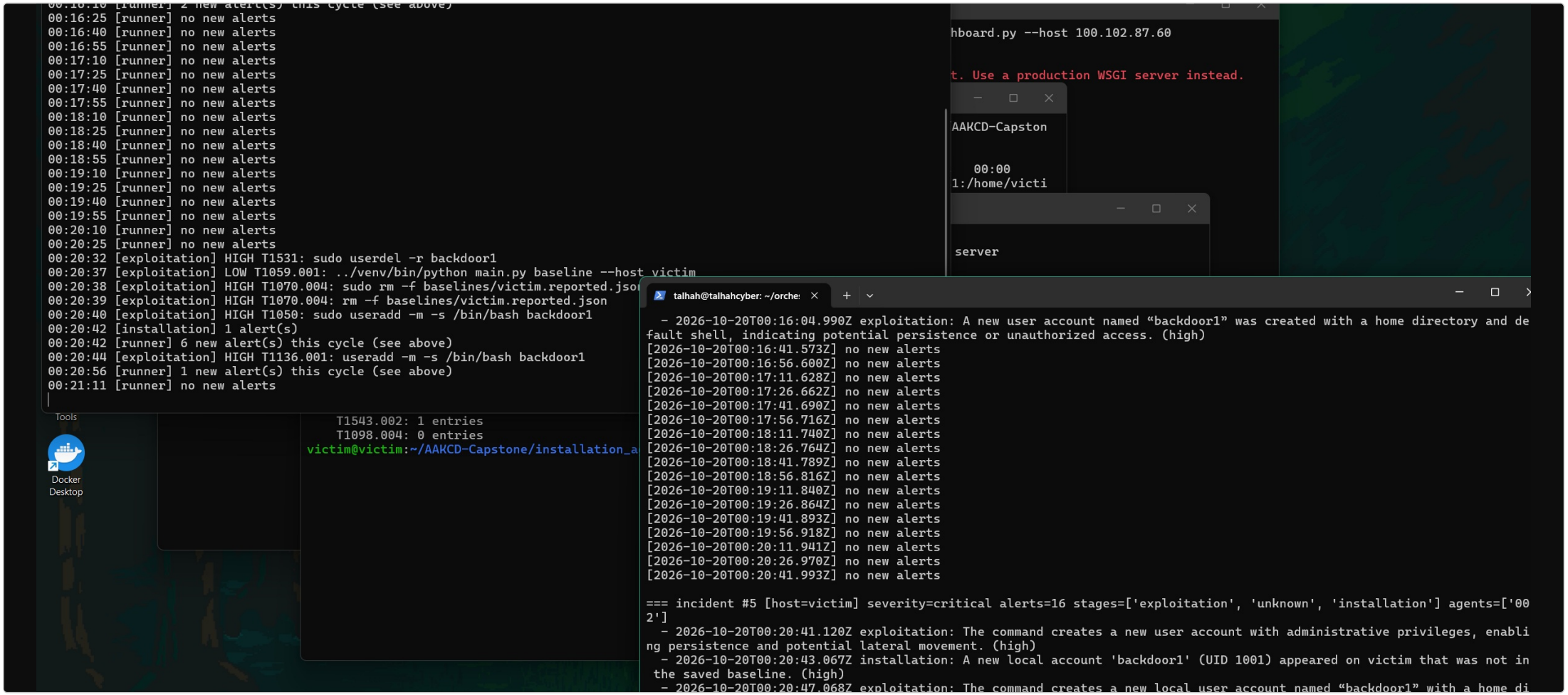
Groq classifies the command as a malicious download and the orchestrator firewall-drops 203.0.113.9.



7e8a1200-image.png

11 SESSION 2 — INSTALLATION — DETECTED + CONTAINED

Stage 4: Installation. A rogue account (backdoor1, UID 1001, T1136.001) appears against the clean baseline; the orchestrator disables it.



ae6aca14-image.png

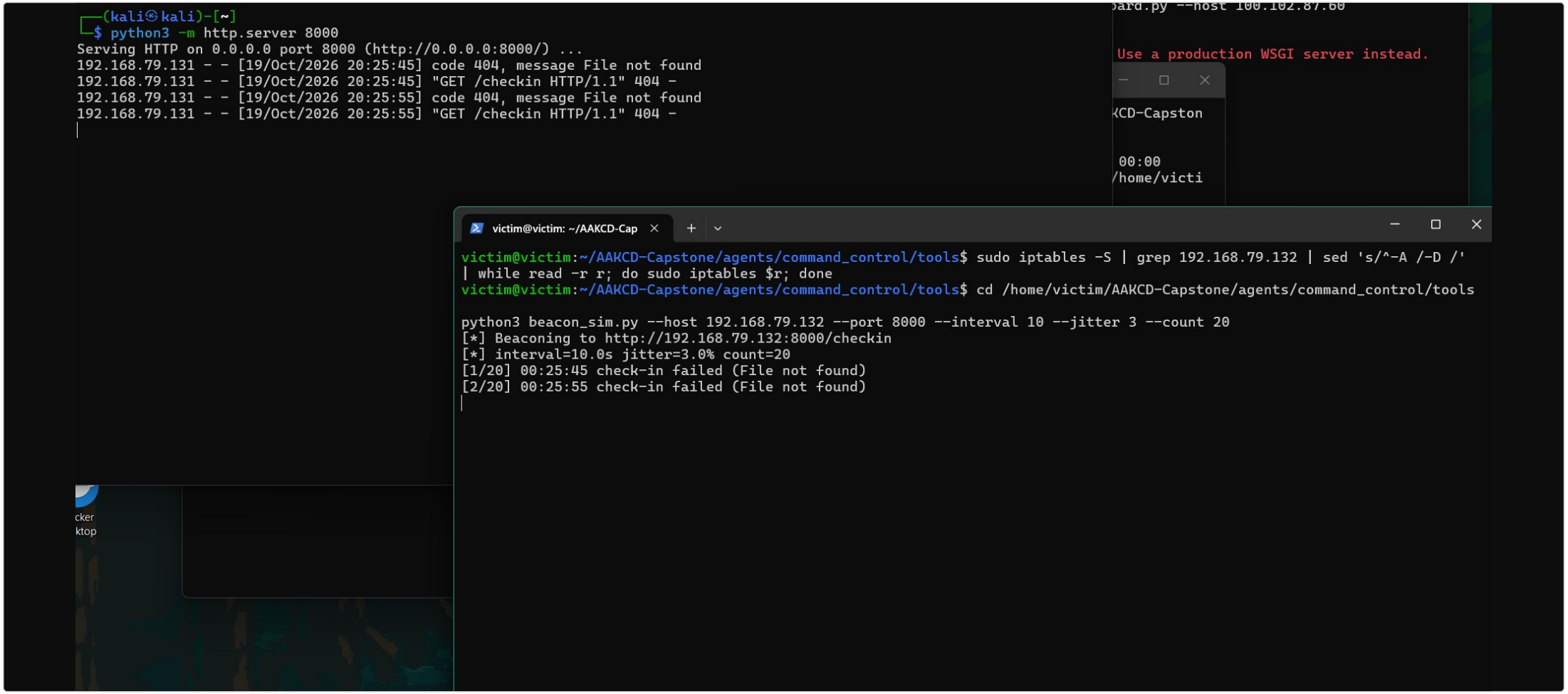
12 SESSION 2 — INSTALLATION — PROOF

passwd -S reports status L: the backdoor account is locked out.

```
victim@victim:~/AAKCD-Capstone/installation_agent$ sudo passwd -S backdoor1
backdoor1 L 2026-10-20 0 99999 7 -1
victim@victim:~/AAKCD-Capstone/installation_agent$
```

f42e652f-image.png

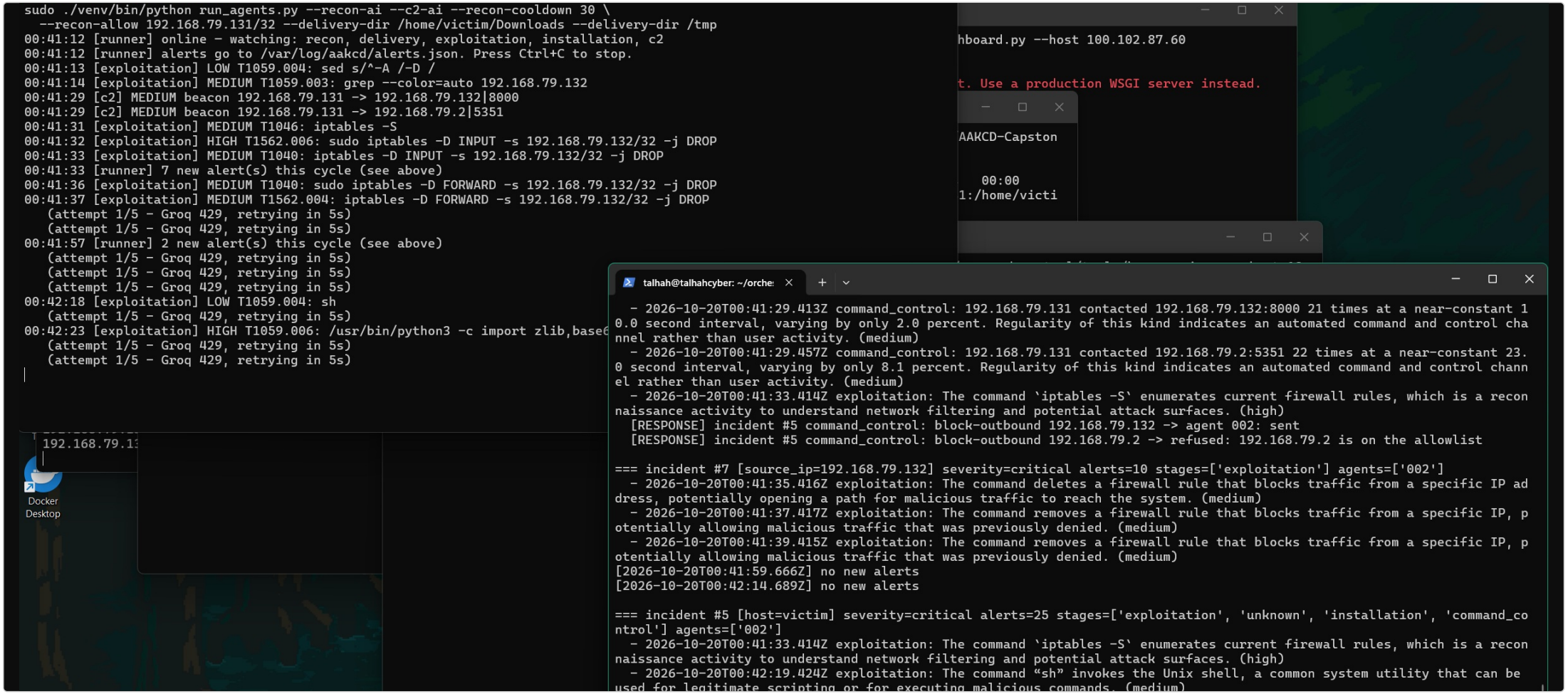
Stage 5: Command & Control. The victim beacons to the attacker's listener every 10 seconds, the classic regular-interval C2 pattern.



f8f662b0-image.png

14 SESSION 3 — COMMAND & CONTROL — DETECTED + CONTAINED

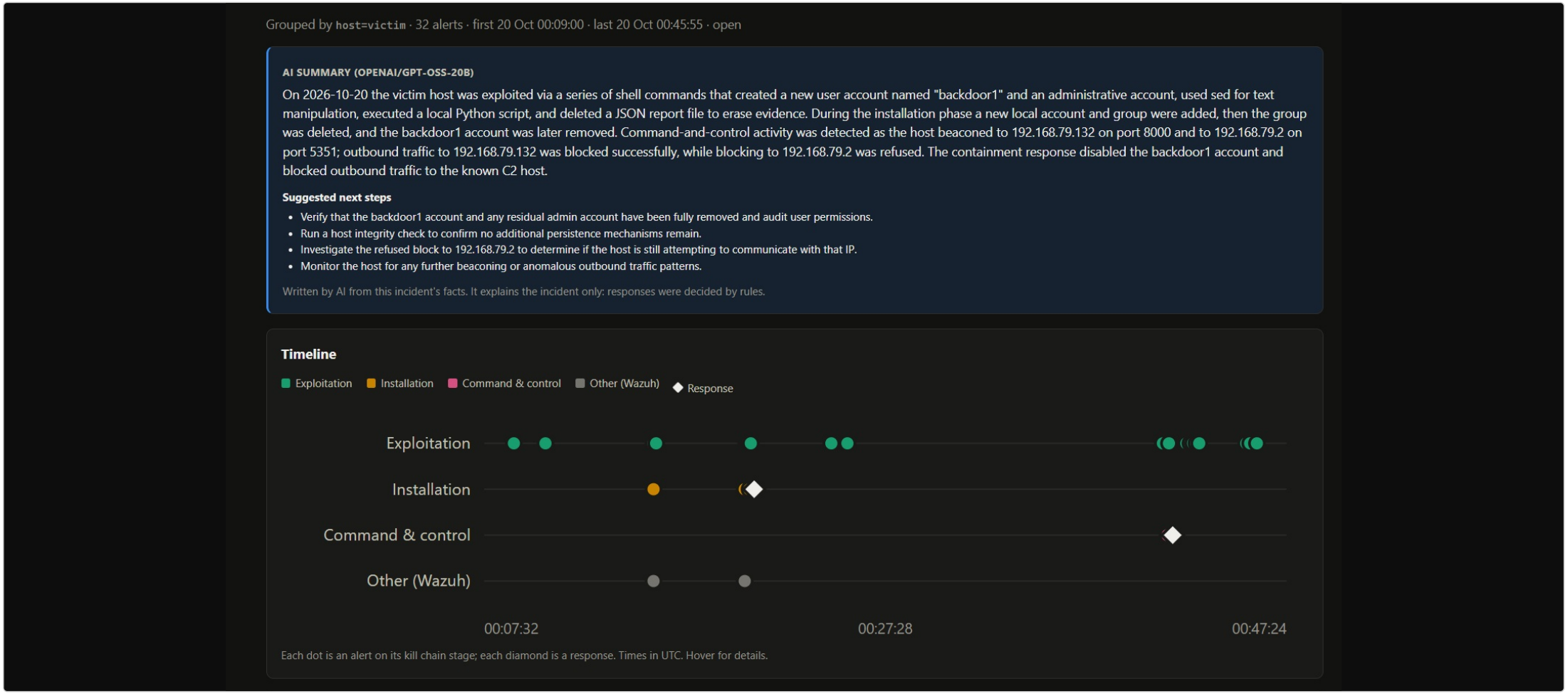
The C2 agent spots the near-constant interval across 21 check-ins and the orchestrator blocks outbound traffic to the server.



b8f7c964-image.png

15 PAYOFF — CORRELATION

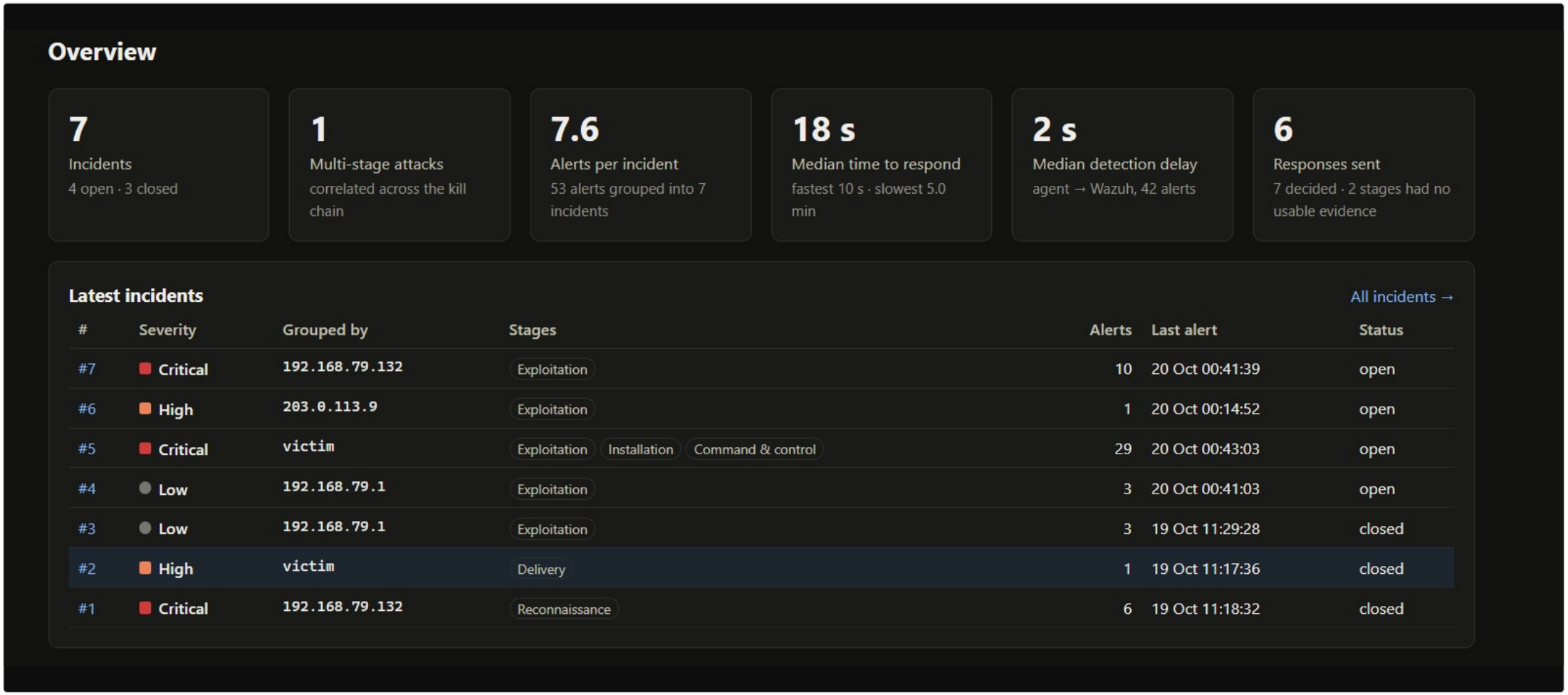
One multi-stage incident: exploitation, installation and C2 on the victim correlated into a single case, with an AI narrative and the responses that contained each stage.



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16 PAYOFF — DASHBOARD OVERVIEW

The whole picture: 7 incidents correlated, 6 responses sent, 18-second median response, one attack tracked across the full kill chain.



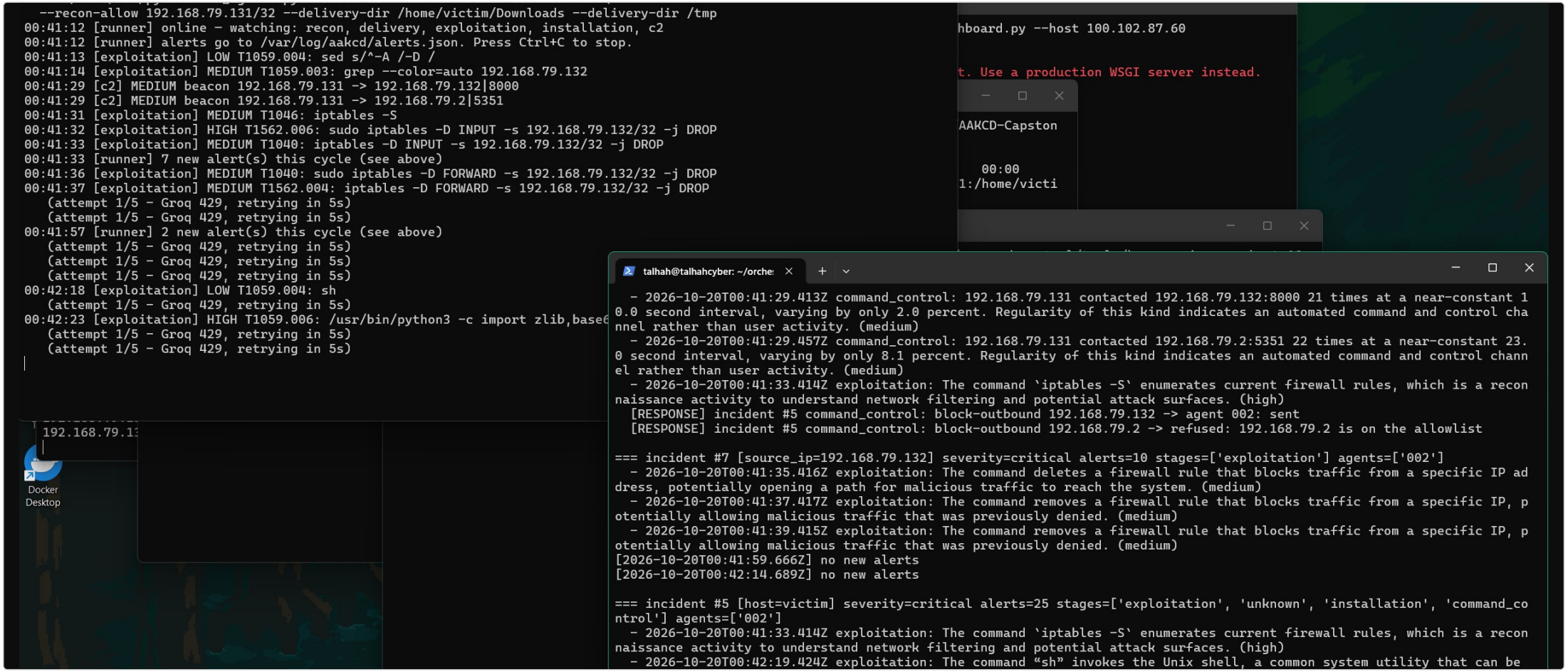
a7270425-image.png

Optional extras

Use these if you want more depth. Skip the Groq-429 frames, incident #7, the older 4-incident dashboard, and the housekeeping shots.

E1 OPTIONAL — SAFETY NET

Same C2 logic also flagged the NAT gateway (192.168.79.2), but the orchestrator refused to block it because it is allowlisted. No self-inflicted outage. (See the 'refused: on the allowlist' line in the poller.)



b8f7c964-image.png

E2 OPTIONAL — RECON AI DETAIL

Each scan alert carries an AI explanation of the activity and the MITRE mapping (T1595).

Time	Stage	Source	Finding	Confidence
19 Oct 11:15:12	Reconnaissance	recon **T1595**	192.168.79.132 probed 20 distinct targets (1 host(s), 20 port(s)) within 60s; 100% of connections were incomplete, consistent with a vertical port scan. **Agent AI:** Between 2026-10-19T11:14:59.730517+00:00 and 2026-10-19T11:14:59.731810+00:00, source IP 192.168.79.132 performed a vertical port scan against a single host, probing 20 distinct ports with 100% incomplete connections. This matches MITRE ATT&CK technique T1595 (Active Scanning) under the Reconnaissance tactic. It matters because it indicates an internal system is actively mapping service availability on a specific target before attempting further exploitation.	high
19 Oct 11:15:14	Reconnaissance	recon **T1595**	192.168.79.132 probed 100 distinct targets (1 host(s), 100 port(s)) within 60s; 100% of connections were incomplete, consistent with a vertical port scan. **Agent AI:** Between 2026-10-19T11:14:59.730517+00:00 and 2026-10-19T11:14:59.735538+00:00, source IP 192.168.79.132 performed a vertical port scan against 100 distinct ports on a single host (192.168.79.131). This activity is mapped to MITRE ATT&CK T1595 (Active Scanning) with a HIGH severity because 100% of the connections attempted during this window were incomplete.	high
19 Oct 11:16:22	Reconnaissance	recon **T1595**	192.168.79.132 probed 20 distinct targets (1 host(s), 20 port(s)) within 60s; 100% of connections were incomplete, consistent with a vertical port scan. **Agent AI:** Between 2026-10-19T11:16:11.457805+00:00 and 2026-10-19T11:16:12.666973+00:00, source IP 192.168.79.132 conducted a vertical port scan against 192.168.79.131. The scanner probed 20 distinct ports within approximately 1.2 seconds, resulting in a 100% incomplete connection ratio. This activity matters because it maps to MITRE ATT&CK T1595 (Active Scanning) and indicates that an internal host is mapping out services on another network node.	high
19 Oct 11:16:24	Reconnaissance	recon **T1595**	192.168.79.132 probed 100 distinct targets (2 host(s), 100 port(s)) within 60s; 99% of connections were incomplete, consistent with a vertical port scan. **Agent AI:** Between 2026-10-19T11:15:18.520370+00:00 and 2026-10-19T11:16:14.288631+00:00, source IP 192.168.79.132 conducted a vertical port scan against 2 distinct hosts, probing 100 distinct ports. This triggered a HIGH severity alert because 99% of the connections were incomplete, which is characteristic of active reconnaissance mapping.	high
19 Oct 11:17:58	Reconnaissance	recon **T1595**	192.168.79.132 probed 1002 distinct targets (3 host(s), 1002 port(s)) within 60s; 100% of connections were incomplete, consistent with a vertical port scan. **Agent AI:** Between 2026-10-19T11:16:10.939951+00:00 and 2026-10-19T11:16:55.261981+00:00, the internal host 192.168.79.132 performed a vertical port scan against 3 distinct hosts across 1002 distinct ports. All 1002 connections were incomplete, which triggered a high-severity reconnaissance alert mapped to MITRE ATT&CK T1595 (Active Scanning). This matters because it indicates active host probing within the network to discover open services.	high
19 Oct 11:18:32	Reconnaissance	recon **T1595**	192.168.79.132 probed 251 distinct targets (2 host(s), 251 port(s)) within 60s; 100% of connections were incomplete, consistent with a vertical port scan. **Agent AI:** Between 2026-10-19T11:16:27.545685+00:00 and 2026-10-19T11:17:27.508218+00:00, source IP 192.168.79.132 conducted a vertical port scan against 2 distinct hosts across 251 distinct ports in a 60-second window. All 251 connections were incomplete, aligning with MITRE ATT&CK technique T1595 (Active Scanning) under the Reconnaissance tactic. This activity matters because it indicates an internal host is actively probing	high

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E3 OPTIONAL — INCIDENT #5 ALERT TIMELINE

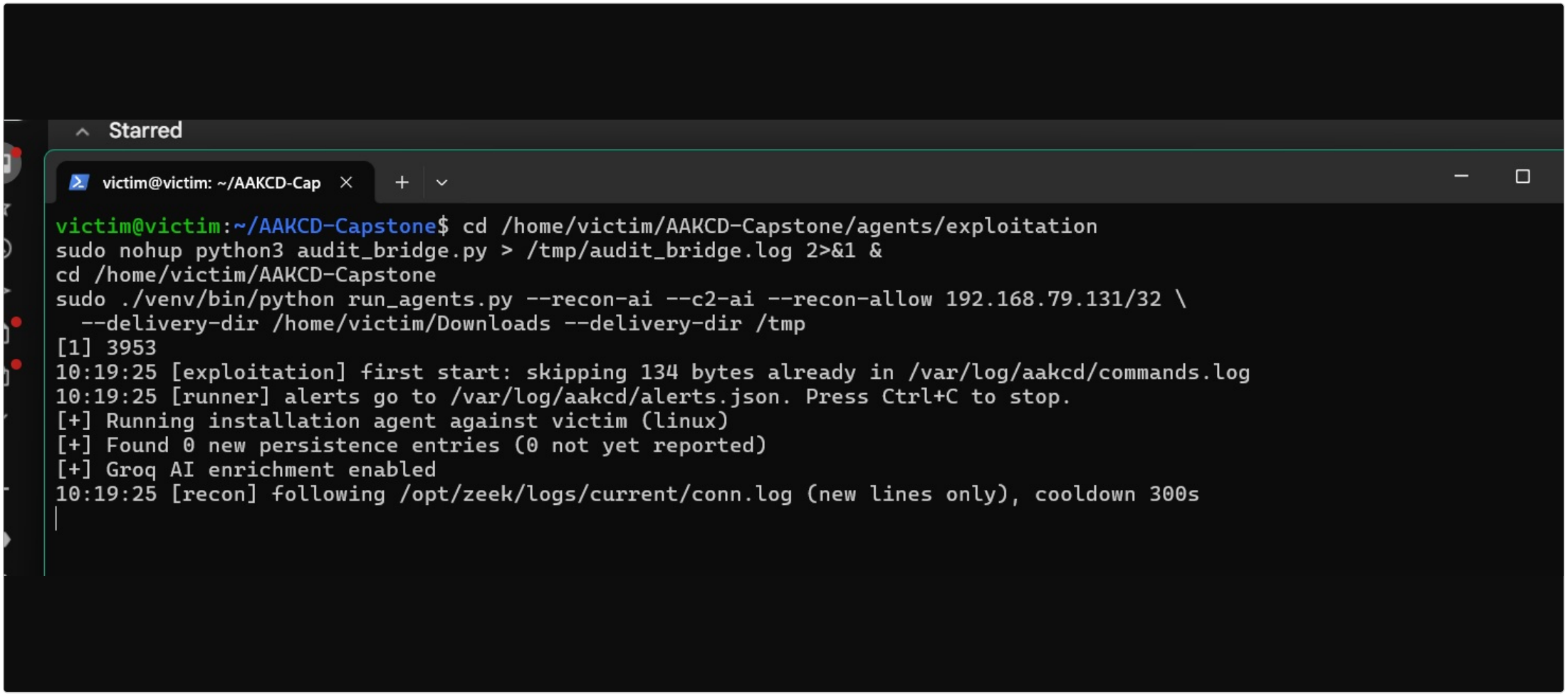
The per-alert AI reasoning behind the multi-stage incident.

20 Oct 00:20:41	Exploitation	exploitation `T1050`	The command creates a new user account with administrative privileges, enabling persistence and potential lateral movement.	high
20 Oct 00:20:43	Installation	installation `T1136.001`	A new local account 'backdoor1' (UID 1001) appeared on victim that was not in the saved baseline. **Agent AI:** Possibly unexpected persistence. A new local account named 'backdoor1' (UID 1001) was created on the host, which was not present in the baseline. The account name and lack of prior existence raise suspicion of unauthorized persistence. Immediate verification of the account's purpose and activity is advised.	high
20 Oct 00:20:47	Exploitation	exploitation `T1136.001`	The command creates a new local user account named "backdoor1" with a home directory and default shell, enabling an attacker to establish persistence and privileged access.	high
20 Oct 00:24:47	Exploitation	exploitation `T1087`	The command queries the password status of the user account "backdoor1," which is a standard account discovery action. It does not modify or exploit the account, so it is considered benign.	high
20 Oct 00:25:33	Exploitation	exploitation `T1059.004`	The command uses `sed` to modify a script or command line, replacing `-A` with `-D`. This is a benign text substitution that could be part of normal configuration or automation.	high
20 Oct 00:25:35	Exploitation	exploitation `T1046`	The command lists current iptables firewall rules, which is a standard administrative task and does not indicate malicious activity.	high
20 Oct 00:41:15	Exploitation	exploitation `T1059.004`	The command uses `sed` to replace `-A` with `-D` in a file, indicating a simple text substitution likely for configuration or script modification.	high
20 Oct 00:41:29	Command & control	command_control `T1071`	192.168.79.131 contacted 192.168.79.132:8000 21 times at a near-constant 10.0 second interval, varying by only 2.0 percent. Regularity of this kind indicates an automated command and control channel rather than user activity. **Agent AI:** Host 192.168.79.131 generated 21 outbound connections to 192.168.79.132 over port 8000 across 3.3 minutes with a near-zero coefficient of variation (0.02) and an average 10-second interval. This highly rigid, regular communication pattern strongly indicates automated polling or command-and-control beaconing rather than human-driven traffic. Because the activity is occurring internally between hosts, it may represent lateral movement, internal staging, or an active implant communicating with an internal listener.	medium
20 Oct 00:41:29	Command & control	command_control `T1071`	192.168.79.131 contacted 192.168.79.2:5351 22 times at a near-constant 23.0 second interval, varying by only 8.1 percent. Regularity of this kind indicates an automated command and control channel rather than user activity. **Agent AI:** Host 192.168.79.131 exhibited highly regular outbound traffic to 192.168.79.2 over port 5351, repeating roughly every 23 seconds across an eight-minute window. The low coefficient of variation in the connection intervals strongly indicates automated beaconing, which is characteristic of command and control activity. Although port 5351 is associated with NAT Port Mapping Protocol (NAT-PMP), the rigid periodicity warrants immediate investigation to rule out covert channel communications.	medium
20 Oct 00:41:33	Exploitation	exploitation `T1046`	The command `iptables -S` enumerates current firewall rules, which is a reconnaissance activity to understand network filtering and potential attack surfaces.	high
20 Oct 00:42:19	Exploitation	exploitation `T1059.004`	The command "sh" invokes the Unix shell, a common system utility that can be used for legitimate scripting or for executing malicious commands.	medium

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E4 OPTIONAL — GO-LIVE DETAIL

A closer look at the startup sequence: the audit bridge backgrounded, then the runner following Zeek's conn.log.



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